

Monish Ramesh Jayakumar

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EDUCATION

Purdue University

Aug. 2025 – May 2028

Bachelor of Science in Computer Science, JMHC Honors, GPA: 3.88

West Lafayette, IN

Coursework: Object-Oriented Programming, Programming in C, Discrete Math, Data Structures and Algorithms, Computer Architecture

TECHNICAL SKILLS

Languages: Python, C, C++, Java, C#, Dart, JavaScript, TypeScript, SQL (PostgreSQL, MySQL)

Frameworks: PyTorch, scikit-learn, DSPy, Flutter, Next.js, React, FastAPI, Unity

Libraries & Tools: NumPy, pandas, Hugging Face, Git, Docker, Linux

EXPERIENCE

Undergraduate Research Assistant

Jan. 2026 – Present

Aphasia Recovery Lab (Dr. Jiyeon Lee)

West Lafayette, IN

- Built a diagnostic pipeline in **PyTorch** and **scikit-learn** separating aphasic speech from control speech at **72%** accuracy
- Built **DSPy** pipelines that tag transcripts with a local LLM and score them in Python, matching clinician scores up to **r = 0.97** correlation
- Extracted linguistic features from **1,600+** clinical speech records using LLM surprisal analysis

Contract Software Developer

May 2026 – July 2026

Hill Mortgage Company

Remote

- Engineered a **Next.js** and **TypeScript** borrower portal on top of a legacy **CodeIgniter/MySQL** backend with document upload and application intake pipelines
- Shipped user profile management **REST** endpoints into the backend, integrating secure session state and extending the legacy production database schema
- Automated **GitHub Actions CI/CD** across dual repositories and refactored responsive UI layouts across **35+** components to eliminate mobile viewport clipping

Founding Engineer

Feb. 2026 – Present

UPlate

West Lafayette, IN

- Co-built a cross-platform **Flutter** app serving **800+** students with real-time dining hall menus and personalized meal recommendations
- Built a scan-to-log pipeline using the **Gemini API** to pull nutrition data from a photo of a meal
- Overhauled Flutter routing architecture by mapping an **86-node** navigation graph, reducing max interaction depth to **2** taps and simplifying redundancies into centralized UI flows

PROJECTS

Campout – Multiplayer Game | C++, FastAPI, PostgreSQL, Docker, Websockets

Jun. 2026 – Present

- Wrote a **C++** server holding a **20 Hz** tick for up to **50** players, with frame-time clamping to absorb lag spikes
- Held **p50** latency under **40ms** with **zero** errors at the full **50-player** cap, load testing a **1-to-50** client sweep with no degradation as concurrency scaled
- Built a **WebSockets** gateway to a **Unix IPC** backend, processing **100-byte** deterministic ticks to fix client drift

Monkish – Chess Engine | Python, PyTorch

Dec. 2025 – May 2026

- Trained a **6-layer** convolutional evaluator on **10 million+** chess positions in **PyTorch**, reaching **r = 0.81** correlation against verified Stockfish evaluations
- Wrote a move generator in **Python** sustaining **140,000+** nodes/sec, verified against **5** standard **perft** positions
- Paired the evaluator with alpha-beta minimax search; drew against a **2000-rated** opponent

MedVR Haptic Glove | C++, C#, Unity, Arduino

Aug. 2024 – Apr. 2025

- Engineered a wearable VR glove tracking fingers to within **2** degrees, feeding **servo-based** touch feedback into a custom **Unity** simulation at up to **15 lbs** of grip resistance
- Invited as the **first** student team to present prototype to engineers at **Thales VisionX**